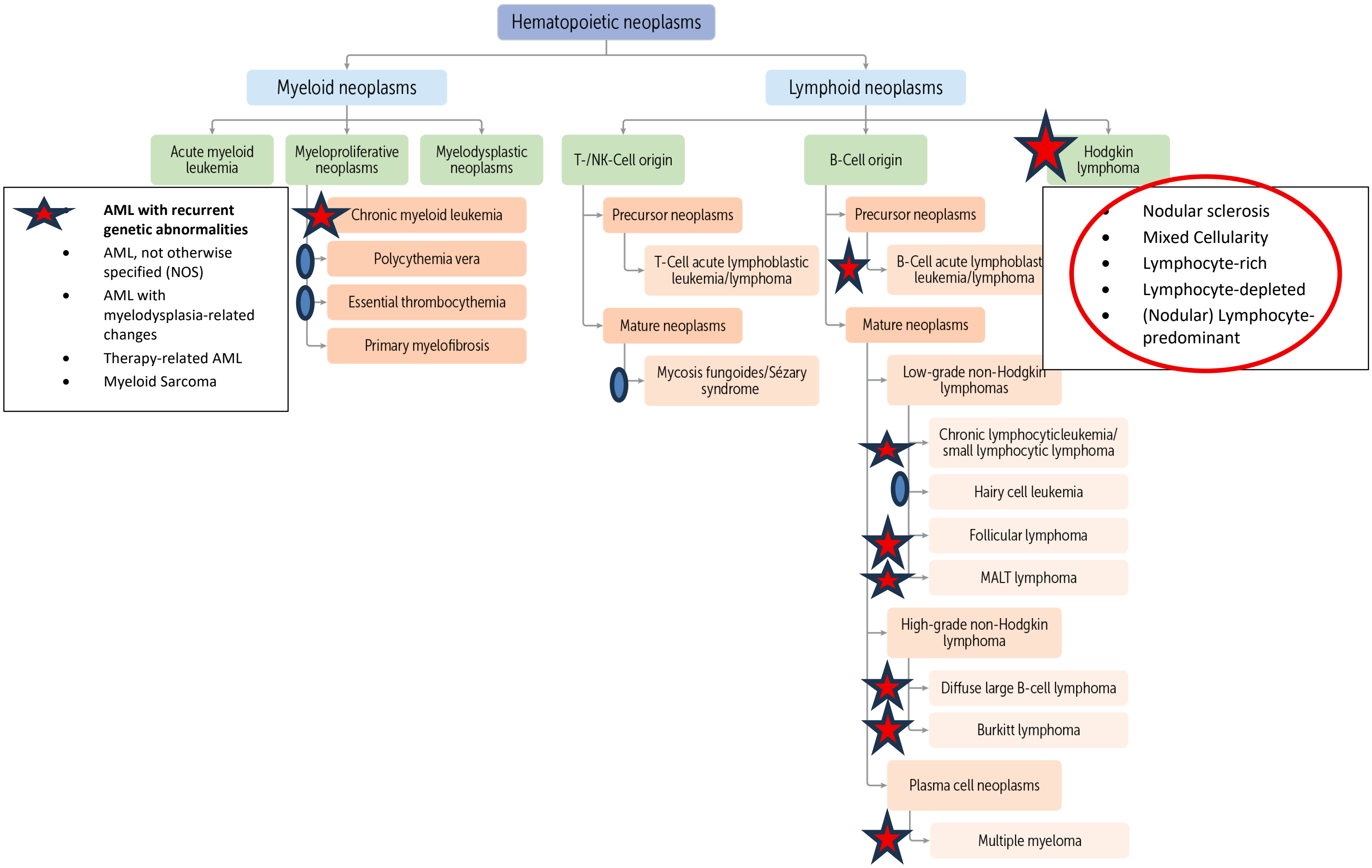

Figure 1 PLEASE NOTE UNDER ACUTE MYELOID LEUKEMIA THERE ARE 5 CATEGORIES BASED ON THE WHO CLASSIFICATION - YOU ARE NOT RESPONSIBLE FOR THE FAB CLASSIFICATION WHICH IS HISTORICAL. ALSO, UNDER HODGKIN LYMPHOMA there are 5 subcategories, the first four are considered Classical.

-Leukemias usually show blood abnormalities, and lymphomas usually present with lymphadenopathy.

-Many of these diseases, especially acute leukemias, multiple myeloma, etc. present similarly because the bone marrow fills with malignant immature cells which out crowd the normal rbcs, platelets, and wbc (myeloid) precursors resulting in anemia (pallor, fatigue), petechiae (low platelets), and infections (the normal wbcs are not present to fight infections)

- **Chronic lymphocytic leukemia/lymphoma (CLL) – B cells**

    - Usually older population (>50-60), presenting with lymphocytosis, lymphadenopathy, hepatosplenomegaly
    - **Smudge cells** on peripheral smear
    - Not many specific translocations
    - **CD19, CD20, CD 23, and CD5**

- **Multiple Myeloma (MM) : plasma cell neoplasm (there are others discussed in Bricks, you are not responsible for at this time)**

    - **monoclonal protein, M spike, seen best on serum electrophoresis (SPEP)**
    - >10% plasma cells, plasmablasts (bone marrow)
    - CRAB: hyper**C**alcemia, Renal dysfunctions, Anemia, Bone lesions)

- Hairy Cell Leukemia – not responsible for but important for Boards; mostly males, usually extensive splenomegaly, "dry" bone marrow tap because the "hairs" on the surface of the cells get enmeshed, historically TRAP positive (immunohistochemical stain), very good prognosis

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- Note: Most lymphomas are B cell lymphomas: Hodgkins v. Non-Hodgkins lymphoma (NHL)
- NHLs include Follicular cell lymphoma, diffuse large B cell lymphoma, Burkitt's lymphoma, Marginal cell lymphoma, Mantle cell lymphoma (all of which come from B cells), and T cell lymphomas (much less common)

- **Follicular Lymphoma** – B cell lymphoma; origin from germinal center cells within follicles of lymph nodes

    - CD markers 19 and 20, and others (but not under 10, those are T cell markers)
    - **Translocation t;(14;18)**
    - **Overexpression of Bcl-2**
    - Usually low grade but 10% may transform to diffuse large B cell lymphoma (which can also happen in CLL)

Diffuse Large B cell Lymphoma (DLBCL) – B cell, much more aggressive, advanced than follicular lymphoma, may be extranodal (outside lymph nodes)

- **Hodgkin lymphoma – tends to spread in contiguous fashion v. non-contiguous for NHL. HL usually has a better prognosis than any NHL**

    - **Reed-Sternberg cells (come from B cells) – diagnostic cell**
    - **Variants of RS cells, EBV associations, CD 15 and CD 30 markers**
    - 5 subtypes of HL; first four **are positive for CD15 and CD30**, but negative for CD20 (and CD45)
    - Nodular sclerosis is the most common subtype; usually good/best prognosis, RS variant known as "lacunar" cell; collagen bands within LN
    - Mixed-cellularity – still good prognosis
    - Lymphocyte-rich – rare; good prognosis, depending on stage
    - Lymphocyte-depleted **– highest association with EBV**
    - Non-classical (nodular) lymphocyte-predominant HL (please note this is different than Lymphocyte-rich) has the **opposite phenotype as the 4 classical ones : CD15 negative and CD30 negative (but positive for CD20 and CD45**) ; usually excellent prognosis; the RS variant is known as the L and H or LP or "popcorn" cell

For all lymphomas, "B symptoms" refer to weight loss, night sweats, fevers, etc.

Review of Important translocations and markers:

- Tdt (lymphoblasts)
- Esterase (Myeloblasts)
- t (15;17) Acute promyelocytic leukemia
- t (9; 22) Philadelphia chromosome CML (minority of cases in Adu;t ALL)
- t (8;14) Burkitt lymphoma
- t(14;18) follicular lymphoma
- CD 15 and CD 30 Hodgkin disease; Classical v. Non-classical – see above
- CD 19, CD20, CD 23, CD 5 Chronic lymphocytic leukemia
- B cell CD markers v. T cell CD markers
- Bcl-2 follicular lymphoma
- Cyclin D-1 (bcl-1) mantle cell lymphoma t(11;14); bcl-1
- Most CD markers less than 10 are T cell markers

Also note that EBV is associated with the **endemic** form of Burkitt's lymphoma and with some forms of Hodgkins lymphomas, especially lymphocyte-depleted

HEMORRHAGE/THROMBOSIS – this is not as detailed as leukemia and lymphoma because nomenclature is clearer and diseases are more straightforward. I've only highlighted major topics. The ScholarRx Bricks for these topics are optional. Please listen to the posted videos I recorded. Be familiar with these, especially:

- Acute v. Chronic Immune thrombocytopenic purpura
- Thrombotic thrombocytopenic purpura (pentad. ADAMST13 deficiency)
- Hemolytic Uremic Syndrome Typical v. Atypical
- Microcytic angiopathic hemolytic anemia (schistocytes)
- Hemophilia A (X-linked inheritance, etc.)
- von Willebrand disease (do not need to know subtypes)
- Disseminated intravascular coagulation (presentation, labs, etc.)
- Factor V Leiden mutation (NOT deficiency) – primary (hereditary) causes of thrombosis
- Lines of Zahn, post mortem clots v. real thombus
- Secondary causes of thrombosis (e.g., prolonged bed rest, obesity, pregnancy, etc.)
- Deep vein thrombosis and pulmonary embolism assocations
- Bernard Soulier v. Glanzmann (platelet adhesion v. aggregation) – these are platelet qualitative disorders
- Important labs including PT/PTT, platelet count, bleeding time, d-dimers, etc.